RESEARCH & CONFERENCE MANAGEMENT SYSTEM

COMPLEX COMPUTING PROBLEM

DATABASE MANAGEMENT SYSTEM

CT-261

SUBMITTED BY:

Your name (Roll No)

Department of Computer Science & Information Technology
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PROJECT OVERVIEW

Introduction:

In today's academic landscape, universities are placing increased emphasis on tracking research output, faculty and student involvement, and academic event participation. To efficiently manage this growing need, an **integrated Research & Conference Management System** is proposed. This system will serve as a centralized platform for managing research activities, publications, and conference participation of faculty and students.

Objective:

The main objective of this project is to design a robust database system that can:

- Track research projects, their contributors, and progress.
- Manage and store metadata related to academic publications.
- Record and monitor participation in academic conferences.
- Provide a unified view of research and academic activities within the university.

Scope of the System:

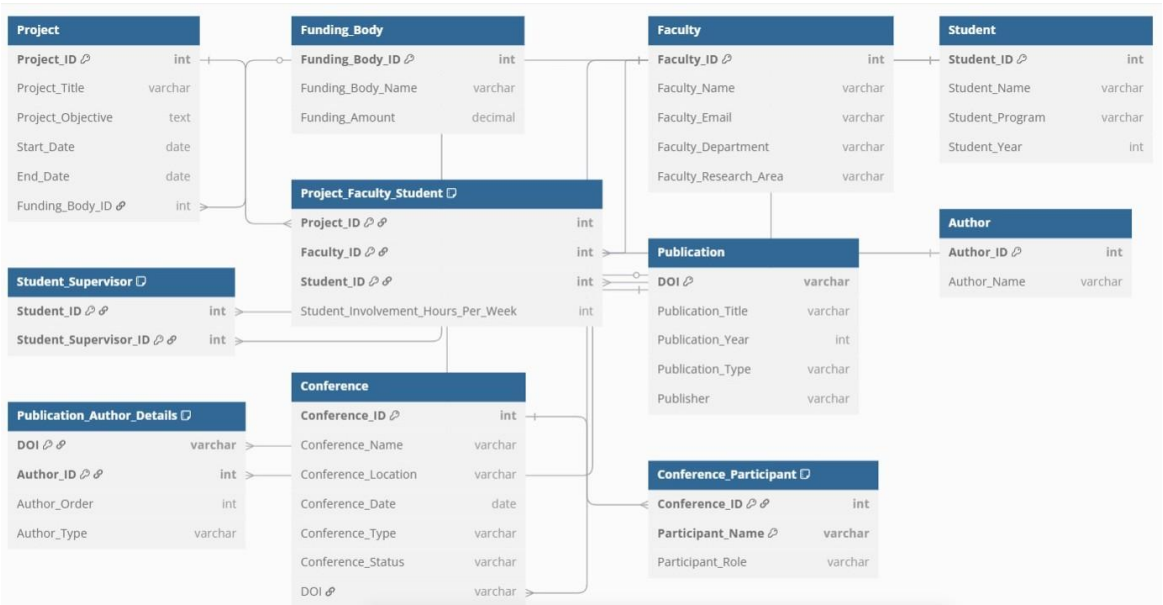
The system will handle the following key modules:

- **Projects**: Each project has details like title, objective, start and end dates, funding information, and participants (faculty and students).
- **Publications**: Includes research papers, book chapters, and conference proceedings. Each publication may have multiple authors, and detailed metadata such as title, year, type, and publisher will be maintained.
- **Conferences**: Tracks both national and international academic events, including data such as type, name, location, date, and participant roles (e.g., presenter, keynote speaker).
- **Faculty and Students**: Stores detailed profiles including name, department, program, and research involvement.

Functional Requirements:

- Track research projects and associated contributors.
- Link publications with authors and associated conferences.
- Identify order of authorship for academic recognition.
- Allow students and faculty to be associated with multiple conferences.
- Ensure integrity constraints such as:
  - A student cannot be involved in more than two projects simultaneously.
  - Each publication must have at least one faculty author.
  - A conference must have at least one publication to be marked as "active"

ENTITY RELATIONSHIP DIAGRAM (ERD):



UNNORMALIZED DATA:

Project_Title	Project_Objective	Start_Date	End_Date
AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024
Quantum Computing Frameworks	Building scalable quantum algorithms	6/1/2021	12/31/2023
Renewable Energy Microgrids	Efficient rural electrification using microgrids	1/1/2023	6/30/2025
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Language Tech for Education	Using NLP to support literacy in rural schools	5/1/2023	12/1/2025

Funding_Amount	Funding_Body	Faculty_Names	Faculty_Emails	Faculty_Departments	Faculty_Research_Areas
50000	National Science Foundation	Dr.Ali, Dr.Ahmed	ali@uni.edu; ahmed@uni.edu	CS; Biomedical	ML; Medical Imaging
750000	Dept. of Science & Tech	Dr. Claire	claire@uni.edu	Physics	Quantum Computing
600000	Green Energy Initiative	Dr. Daniel	daniel@uni.edu	Electrical	Renewable Energy
400000	Tech Robotics Foundation	Dr. Eva; Dr. Alice	eva@uni.edu; alice@uni.edu	Robotics; CS	HRI; Cognitive Systems
350000	UNESCO	Dr. Farah	farah@uni.edu	Linguistics	NLP

Student_Names	Student_Programs	Student_Years	Student_Supervisors	Student_Involvement_Hours
Hamza Malik; Asad Jamal	PhD; MSc	2; 1	Dr.Ali; Dr.Ahmed	10; 15
Alex Zhang	MSc	1	Dr. Claire	20
Riya Singh	PhD	3	Dr. Daniel	25
Omar Khan	MSc	2	Dr. Eva; Dr. Alice	12
Abdul Ahad; Huzaifa Khan	PhD; MSc	2; 1	Dr. Farah	18; 10

Publication_Titles	Publication_Years	Publication_types	Publishers	DOIs
Deep learning for X-ray; ML in cardiology	2023; 2024	Conference Paper; Book Chapter	Springer; IEEE	10.1000/xyz1; 10.1000/xyz2
Quantum Entanglement in Logic Gates	2023	Journal Article	Elsevier	10.1000/xyz3
Smart Microgrids for India	2024	Conference Paper	IEEE	10.1000/xyz4
Gesture Recognition in HRI	2024	Book Chapter	Springer	10.1000/xyz5
NLP for Primary Learning; Speech Tools for Kids	2023; 2024	Journal Article; Conference Paper	Wiley; ACM	10.1000/xyz6; 10.1000/xyz7

Authors_Order	Conference_Names	Conference_Locations	Conference_Dates
Hamza,Dr.Ali; Asad,Dr.Ahmed	Int.Conference on AI; Nat. MedTech Meet	Paris,France; Islamabad, Pakistan	6/15/2023; 2/20/2024
Alex Zhang, Dr.Claire	Int. Quantum Symposium	Geneva	9/10/2023
Riya Singh, Dr. Daniel	Nat. Energy Innovation Summit	Mumbai	3/12/2024
Dr. Eva,Omar Khan,Dr. Alice	Robotic Futures Conference	Tokyo	5/20/2024
Abdul Ahad, Dr.Farah; Huzaifa Kahn, Dr.Farah	EdTech Global Forum	London	8/10/2024

Conference_Types	Participants_Names	Participants_Roles
International; National	Dr.Ali; hamza; Asad	Keynote Speaker;Presenter; Attendee
International	Dr. Claire; Alex Zhang	Presenter; Attendee
National	Dr. Daniel; Riya Singh	Attendee; Presenter
International	Dr. Eva; Omar Khan; Dr. Alice	Presenter; Presenter; Panelist
International	Dr. Farah; Abdul Ahad; Huzaifa Khan	Keynote Speaker; Presenter; Attendee

This table is not in 1NF as it contains multi-valued atrinubutes i.e. Faculty_Names, Faculty_Emails, Faculty_Departments, Faculty_Research_Areas, Student_Names, Student_Programs, Student_Years, Student_Supervisors, Student_Involvement_Hours, Publication_Titles, Publication_Years, Pubications_Types, Publishers, DOIs, Author_Orders, Conference_Names, Conferenece_Locations, Conference_Dates, Conference_types, Participants_Names, Participants_roles.

*these are the columns of same unnormalized table.

ANOMALIES:

1. Insertion Anomaly:

An insertion anomaly occurs when the database design requires the presence of unnecessary or unrelated data in order to insert a valid piece of information.

Example: If a newly hired faculty member (e.g., *Dr. Noor*) has not yet been assigned to any project, their information (such as ID, name, email, department, and research areas) cannot be added to the system unless a corresponding project and its funding details are also entered. This limits the ability to store independent faculty records.

Funding_Amount	Funding_Body	Faculty_Names	Faculty_Emails	Faculty_Departments	Faculty_Research_Areas
50000	National Science Foundation	Dr.Ali, Dr.Ahmed	ali@uni.edu; ahmed@uni.edu	CS; Biomedical	ML; Medical Imaging
750000	Dept. of Science & Tech	Dr. Claire	claire@uni.edu	Physics	Quantum Computing
600000	Green Energy Initiative	Dr. Daniel	daniel@uni.edu	Electrical	Renewable Energy
400000	Tech Robotics Foundation	Dr. Eva; Dr. Alice	eva@uni.edu; alice@uni.edu	Robotics; CS	HRI; Cognitive Systems
350000	UNESCO	Dr. Farah	farah@uni.edu	Linguistics	NLP
		Dr. Noor	noor@uni.com	Information Technology	Blockchain Technology

Insertion Anomaly

2. Deletion Anomaly:

A deletion anomaly occurs when the removal of a single record inadvertently results in the loss of additional, valuable information.

Example: If a student is the sole contributor to a project (e.g., *Alex Zhang* in "Quantum Computing Frameworks") and their data is deleted, all related project information—such as title, objective, funding, and associated faculty—would also be lost due to the structure of the unnormalized table.

Project_Title	Project_Objective	Student_Names	Student_Programs	Student_Years
AI in Healthcare	Improving Diagnostic with AI	Hamza Malik; Asad Jamal	PhD; MSc	2; 1
Quantum Computing Frameworks	Building scalable quantum algorithms	Alex Zhang	MSc	1
Renewable Energy Microgrids	Efficient rural electrification using microgrids	Riya Singh	PhD	3
Human-Robot Interaction	Enhancing collaboration between humans and robots	Omar Khan	MSc	2
Language Tech for Education	Using NLP to support literacy in rural schools	Abdul Ahad; Huzaifa Khan	PhD; MSc	2; 1

Deletion Anomaly

3. Updation Anomaly:
An update anomaly occurs when redundant data exists in a database, and updating one instance of the data requires updating all others to maintain consistency.

Example: If a faculty member's email (e.g., *Dr. Claire's* email) is changed, this update must be made in every occurrence where that faculty appears. Failure to update all instances may result in inconsistent contact information.

Faculty_Names	Faculty_Emails	Faculty_Departments	Faculty_Research_Areas	Student_Names
Dr.Ali, Dr.Ahmed	ali@uni.edu; ahmed@uni.edu	CS; Biomedical	ML; Medical Imaging	Hamza Malik; Asad Jamal
Dr. Claire	claire@uni.edu	Physics	Quantum Computing	Alex Zhang
Dr. Daniel	daniel@uni.edu	Electrical	Renewable Energy	Riya Singh
Dr. Eva; Dr. Alice	eva@uni.edu; alice@uni.edu	Robotics; CS	HRI; Cognitive Systems	Omar Khan
Dr. Claire	claire@uni.edu	Physics	Quantum Computing	Abdul Ahad; Huzaifa Khan

NORMALIZATION:

Normalization minimizes data redundancy and prevents anomalies. It helps maintain data consistency, improves integrity, and ensures the database remains efficient and easy to manage.

Normalization is achieved by applying normal forms step by step—starting from 1NF to remove repeating groups, 2NF to eliminate partial dependencies, and 3NF to remove transitive dependencies. This process ensures a structured and consistent database.

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1NF:

Unnormalized table is converted into 1NF as follows:

Project_Title	Project_Objective	Start_Date	End_Date
AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024
AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024
AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024
AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024
AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024
AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024
AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024
AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024
AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024
AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024
AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024
AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024
Quantum Computing Frameworks	Building scalable quantum algorithms	6/1/2021	12/31/2023
Quantum Computing Frameworks	Building scalable quantum algorithms	6/1/2021	12/31/2023
Quantum Computing Frameworks	Building scalable quantum algorithms	6/1/2021	12/31/2023
Quantum Computing Frameworks	Building scalable quantum algorithms	6/1/2021	12/31/2023
Renewable Energy Microgrids	Efficient rural electrification using microgrids	1/1/2023	6/30/2025
Renewable Energy Microgrids	Efficient rural electrification using microgrids	1/1/2023	6/30/2025
Renewable Energy Microgrids	Efficient rural electrification using microgrids	1/1/2023	6/30/2025
Renewable Energy Microgrids	Efficient rural electrification using microgrids	1/1/2023	6/30/2025
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
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Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024
Language Tech for Education	Using NLP to support literacy in rural schools	5/1/2023	12/1/2025
Language Tech for Education	Using NLP to support literacy in rural schools	5/1/2023	12/1/2025
Language Tech for Education	Using NLP to support literacy in rural schools	5/1/2023	12/1/2025
Language Tech for Education	Using NLP to support literacy in rural schools	5/1/2023	12/1/2025
Language Tech for Education	Using NLP to support literacy in rural schools	5/1/2023	12/1/2025
Language Tech for Education	Using NLP to support literacy in rural schools	5/1/2023	12/1/2025
Language Tech for Education	Using NLP to support literacy in rural schools	5/1/2023	12/1/2025
Language Tech for Education	Using NLP to support literacy in rural schools	5/1/2023	12/1/2025
Language Tech for Education	Using NLP to support literacy in rural schools	5/1/2023	12/1/2025
Language Tech for Education	Using NLP to support literacy in rural schools	5/1/2023	12/1/2025
Language Tech for Education	Using NLP to support literacy in rural schools	5/1/2023	12/1/2025

continued

Funding_Amount	Funding_Body	Faculty_Names	Faculty_Emails	Faculty_Departments
50000	National Science Foundation	Dr.Ali	ali@uni.edu	CS
50000	National Science Foundation	Dr.Ali	ali@uni.edu	CS
50000	National Science Foundation	Dr.Ali	ali@uni.edu	CS
50000	National Science Foundation	Dr.Ali	ali@uni.edu	CS
50000	National Science Foundation	Dr.Ali	ali@uni.edu	CS
50000	National Science Foundation	Dr.Ali	ali@uni.edu	CS
50000	National Science Foundation	Dr.Ahmed	ahmed@uni.edu	Biomedical
50000	National Science Foundation	Dr.Ahmed	ahmed@uni.edu	Biomedical
50000	National Science Foundation	Dr.Ahmed	ahmed@uni.edu	Biomedical
50000	National Science Foundation	Dr.Ahmed	ahmed@uni.edu	Biomedical
50000	National Science Foundation	Dr.Ahmed	ahmed@uni.edu	Biomedical
50000	National Science Foundation	Dr.Ahmed	ahmed@uni.edu	Biomedical
750000	Dept. of Science & Tech	Dr. Claire	claire@uni.edu	Physics
750000	Dept. of Science & Tech	Dr. Claire	claire@uni.edu	Physics
750000	Dept. of Science & Tech	Dr. Claire	claire@uni.edu	Physics
750000	Dept. of Science & Tech	Dr. Claire	claire@uni.edu	Physics
600000	Green Energy Initiative	Dr. Daniel	daniel@uni.edu	Electrical
600000	Green Energy Initiative	Dr. Daniel	daniel@uni.edu	Electrical
600000	Green Energy Initiative	Dr. Daniel	daniel@uni.edu	Electrical
600000	Green Energy Initiative	Dr. Daniel	daniel@uni.edu	Electrical
400000	Tech Robotics Foundation	Dr. Eva	eva@uni.edu	Robotics
400000	Tech Robotics Foundation	Dr. Eva	eva@uni.edu	Robotics
400000	Tech Robotics Foundation	Dr. Eva	eva@uni.edu	Robotics
400000	Tech Robotics Foundation	Dr. Eva	eva@uni.edu	Robotics
400000	Tech Robotics Foundation	Dr. Eva	eva@uni.edu	Robotics
400000	Tech Robotics Foundation	Dr. Eva	eva@uni.edu	Robotics
400000	Tech Robotics Foundation	Dr. Eva	eva@uni.edu	Robotics
400000	Tech Robotics Foundation	Dr. Eva	eva@uni.edu	Robotics
400000	Tech Robotics Foundation	Dr. Eva	eva@uni.edu	Robotics
400000	Tech Robotics Foundation	Dr. Alice	alice@uni.edu	CS
400000	Tech Robotics Foundation	Dr. Alice	alice@uni.edu	CS
400000	Tech Robotics Foundation	Dr. Alice	alice@uni.edu	CS
400000	Tech Robotics Foundation	Dr. Alice	alice@uni.edu	CS
400000	Tech Robotics Foundation	Dr. Alice	alice@uni.edu	CS
400000	Tech Robotics Foundation	Dr. Alice	alice@uni.edu	CS
400000	Tech Robotics Foundation	Dr. Alice	alice@uni.edu	CS
400000	Tech Robotics Foundation	Dr. Alice	alice@uni.edu	CS
400000	Tech Robotics Foundation	Dr. Alice	alice@uni.edu	CS
350000	UNESCO	Dr. Farah	farah@uni.edu	Linguistics
350000	UNESCO	Dr. Farah	farah@uni.edu	Linguistics
350000	UNESCO	Dr. Farah	farah@uni.edu	Linguistics
350000	UNESCO	Dr. Farah	farah@uni.edu	Linguistics
350000	UNESCO	Dr. Farah	farah@uni.edu	Linguistics
350000	UNESCO	Dr. Farah	farah@uni.edu	Linguistics
350000	UNESCO	Dr. Farah	farah@uni.edu	Linguistics
350000	UNESCO	Dr. Farah	farah@uni.edu	Linguistics
350000	UNESCO	Dr. Farah	farah@uni.edu	Linguistics
350000	UNESCO	Dr. Farah	farah@uni.edu	Linguistics
350000	UNESCO	Dr. Farah	farah@uni.edu	Linguistics

continued

Faculty_Research_Areas	Student_Names	Student_Programs	Student_Years	Student_Supervisors
ML	Hamza Malik	PhD	2	Dr.Ali
ML	Hamza Malik	PhD	2	Dr.Ali
ML	Hamza Malik	PhD	2	Dr.Ali
ML	Hamza Malik	PhD	2	Dr.Ali
ML	Hamza Malik	PhD	2	Dr.Ali
ML	Hamza Malik	PhD	2	Dr.Ali
Medical Imaging	Asad Jamal	MSc	1	Dr.Ahmed
Medical Imaging	Asad Jamal	MSc	1	Dr.Ahmed
Medical Imaging	Asad Jamal	MSc	1	Dr.Ahmed
Medical Imaging	Asad Jamal	MSc	1	Dr.Ahmed
Medical Imaging	Asad Jamal	MSc	1	Dr.Ahmed
Medical Imaging	Asad Jamal	MSc	1	Dr.Ahmed
Quantum Computing	Alex Zhang	MSc	1	Dr. Claire
Quantum Computing	Alex Zhang	MSc	1	Dr. Claire
Quantum Computing	Alex Zhang	MSc	1	Dr. Claire
Quantum Computing	Alex Zhang	MSc	1	Dr. Claire
Renewable Energy	Riya Singh	PhD	3	Dr. Daniel
Renewable Energy	Riya Singh	PhD	3	Dr. Daniel
Renewable Energy	Riya Singh	PhD	3	Dr. Daniel
Renewable Energy	Riya Singh	PhD	3	Dr. Daniel
HRI	Omar Khan	MSc	2	Dr. Eva
HRI	Omar Khan	MSc	2	Dr. Eva
HRI	Omar Khan	MSc	2	Dr. Eva
HRI	Omar Khan	MSc	2	Dr. Eva
HRI	Omar Khan	MSc	2	Dr. Eva
HRI	Omar Khan	MSc	2	Dr. Eva
HRI	Omar Khan	MSc	2	Dr. Eva
HRI	Omar Khan	MSc	2	Dr. Eva
HRI	Omar Khan	MSc	2	Dr. Eva
Cognitive Systems	Omar Khan	MSc	2	Dr. Alice
Cognitive Systems	Omar Khan	MSc	2	Dr. Alice
Cognitive Systems	Omar Khan	MSc	2	Dr. Alice
Cognitive Systems	Omar Khan	MSc	2	Dr. Alice
Cognitive Systems	Omar Khan	MSc	2	Dr. Alice
Cognitive Systems	Omar Khan	MSc	2	Dr. Alice
Cognitive Systems	Omar Khan	MSc	2	Dr. Alice
Cognitive Systems	Omar Khan	MSc	2	Dr. Alice
Cognitive Systems	Omar Khan	MSc	2	Dr. Alice
NLP	Abdul Ahad	PhD	2	Dr. Farah
NLP	Abdul Ahad	PhD	2	Dr. Farah
NLP	Abdul Ahad	PhD	2	Dr. Farah
NLP	Abdul Ahad	PhD	2	Dr. Farah
NLP	Abdul Ahad	PhD	2	Dr. Farah
NLP	Abdul Ahad	PhD	2	Dr. Farah
NLP	Huzaifa Khan	MSc	1	Dr. Farah
NLP	Huzaifa Khan	MSc	1	Dr. Farah
NLP	Huzaifa Khan	MSc	1	Dr. Farah
NLP	Huzaifa Khan	MSc	1	Dr. Farah
NLP	Huzaifa Khan	MSc	1	Dr. Farah
NLP	Huzaifa Khan	MSc	1	Dr. Farah

continued

Student_Involvement_Hours	Publication_Titles	Publication_Years	Publication_types
10	Deep learning for X-ray	2023	Conference Paper
10	Deep learning for X-ray	2023	Conference Paper
10	Deep learning for X-ray	2023	Conference Paper
10	Deep learning for X-ray	2023	Conference Paper
10	Deep learning for X-ray	2023	Conference Paper
10	Deep learning for X-ray	2023	Conference Paper
15	ML in cardiology	2024	Book Chapter
15	ML in cardiology	2024	Book Chapter
15	ML in cardiology	2024	Book Chapter
15	ML in cardiology	2024	Book Chapter
15	ML in cardiology	2024	Book Chapter
15	ML in cardiology	2024	Book Chapter
20	Quantum Entanglement in Logic Gates	2023	Journal Article
20	Quantum Entanglement in Logic Gates	2023	Journal Article
20	Quantum Entanglement in Logic Gates	2023	Journal Article
20	Quantum Entanglement in Logic Gates	2023	Journal Article
25	Smart Microgrids for India	2024	Conference Paper
25	Smart Microgrids for India	2024	Conference Paper
25	Smart Microgrids for India	2024	Conference Paper
25	Smart Microgrids for India	2024	Conference Paper
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
12	Gesture Recognition in HRI	2024	Book Chapter
18	NLP for Primary Learning	2023	Journal Article
18	NLP for Primary Learning	2023	Journal Article
18	NLP for Primary Learning	2023	Journal Article
18	NLP for Primary Learning	2023	Journal Article
18	NLP for Primary Learning	2023	Journal Article
18	NLP for Primary Learning	2023	Journal Article
10	Speech Tools for Kids	2024	Conference Paper
10	Speech Tools for Kids	2024	Conference Paper
10	Speech Tools for Kids	2024	Conference Paper
10	Speech Tools for Kids	2024	Conference Paper
10	Speech Tools for Kids	2024	Conference Paper
10	Speech Tools for Kids	2024	Conference Paper

continued

Publishers	DOIs	Authors_Names	Authors_Orders	Conference_Names
Springer	10.1000/xyz1	Hamza	1	Int.Conference on AI
Springer	10.1000/xyz1	Dr.Ali	2	Int.Conference on AI
Springer	10.1000/xyz1	Hamza	1	Int.Conference on AI
Springer	10.1000/xyz1	Dr.Ali	2	Int.Conference on AI
Springer	10.1000/xyz1	Hamza	1	Int.Conference on AI
Springer	10.1000/xyz1	Dr.Ali	2	Int.Conference on AI
IEEE	10.1000/xyz2	Asad	1	Nat. MedTech Meet
IEEE	10.1000/xyz2	Dr.Ahmed	2	Nat. MedTech Meet
IEEE	10.1000/xyz2	Asad	1	Nat. MedTech Meet
IEEE	10.1000/xyz2	Dr.Ahmed	2	Nat. MedTech Meet
IEEE	10.1000/xyz2	Asad	1	Nat. MedTech Meet
IEEE	10.1000/xyz2	Dr.Ahmed	2	Nat. MedTech Meet
Elsevier	10.1000/xyz3	Alex Zhang	1	Int. Quantum Symposium
Elsevier	10.1000/xyz3	Dr.Clarie	2	Int. Quantum Symposium
Elsevier	10.1000/xyz3	Alex Zhang	1	Int. Quantum Symposium
Elsevier	10.1000/xyz3	Dr.Clarie	2	Int. Quantum Symposium
IEEE	10.1000/xyz4	Riya Singh	1	Nat. Energy Innovation Summit
IEEE	10.1000/xyz4	Dr. Daniel	2	Nat. Energy Innovation Summit
IEEE	10.1000/xyz4	Riya Singh	1	Nat. Energy Innovation Summit
IEEE	10.1000/xyz4	Daniel	2	Nat. Energy Innovation Summit
Springer	10.1000/xyz5	Dr. Eva	1	Robotic Futures Conference
Springer	10.1000/xyz5	Omar Khan	2	Robotic Futures Conference
Springer	10.1000/xyz5	Dr. Alice	3	Robotic Futures Conference
Springer	10.1000/xyz5	Dr. Eva	1	Robotic Futures Conference
Springer	10.1000/xyz5	Omar Khan	2	Robotic Futures Conference
Springer	10.1000/xyz5	Dr. Alice	3	Robotic Futures Conference
Springer	10.1000/xyz5	Dr. Eva	1	Robotic Futures Conference
Springer	10.1000/xyz5	Omar Khan	2	Robotic Futures Conference
Springer	10.1000/xyz5	Dr. Alice	3	Robotic Futures Conference
Springer	10.1000/xyz5	Dr. Eva	1	Robotic Futures Conference
Springer	10.1000/xyz5	Omar Khan	2	Robotic Futures Conference
Springer	10.1000/xyz5	Dr. Alice	3	Robotic Futures Conference
Springer	10.1000/xyz5	Dr. Eva	1	Robotic Futures Conference
Springer	10.1000/xyz5	Omar Khan	2	Robotic Futures Conference
Springer	10.1000/xyz5	Dr. Alice	3	Robotic Futures Conference
Springer	10.1000/xyz5	Dr. Eva	1	Robotic Futures Conference
Springer	10.1000/xyz5	Omar Khan	2	Robotic Futures Conference
Springer	10.1000/xyz5	Dr. Alice	3	Robotic Futures Conference
Wiley	10.1000/xyz6	Abdul Ahad	1	EdTech Global Forum
Wiley	10.1000/xyz6	Dr.Farah	2	EdTech Global Forum
Wiley	10.1000/xyz6	Abdul Ahad	1	EdTech Global Forum
Wiley	10.1000/xyz6	Dr.Farah	2	EdTech Global Forum
Wiley	10.1000/xyz6	Abdul Ahad	1	EdTech Global Forum
Wiley	10.1000/xyz6	Dr.Farah	2	EdTech Global Forum
ACM	10.1000/xyz7	Huzaifa Khan	1	EdTech Global Forum
ACM	10.1000/xyz7	Dr.Farah	2	EdTech Global Forum
ACM	10.1000/xyz7	Huzaifa Khan	1	EdTech Global Forum
ACM	10.1000/xyz7	Dr.Farah	2	EdTech Global Forum
ACM	10.1000/xyz7	Huzaifa Khan	1	EdTech Global Forum
ACM	10.1000/xyz7	Dr.Farah	2	EdTech Global Forum

continued

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Conference_Locations	Conference_Dates	Conference_Types	Participants_Names	Participants_Roles
Paris,France	6/15/2023	International	Dr. Ali	Keynote Speaker
Paris,France	6/15/2023	International	Dr. Ali	Keynote Speaker
Paris,France	6/15/2023	International	Hamza	Presenter
Paris,France	6/15/2023	International	Hamza	Presenter
Paris,France	6/15/2023	International	Asad	Attendee
Paris,France	6/15/2023	International	Asad	Attendee
Islamabad, Pakistan	2/20/2024	National	Dr. Ali	Keynote Speaker
Islamabad, Pakistan	2/20/2024	National	Dr. Ali	Keynote Speaker
Islamabad, Pakistan	2/20/2024	National	Hamza	Presenter
Islamabad, Pakistan	2/20/2024	National	Hamza	Presenter
Islamabad, Pakistan	2/20/2024	National	Asad	Attendee
Islamabad, Pakistan	2/20/2024	National	Asad	Attendee
Geneva	9/10/2023	International	Dr. Claire	Presenter
Geneva	9/10/2023	International	Dr. Claire	Presenter
Geneva	9/10/2023	International	Alex Zhang	Attendee
Geneva	9/10/2023	International	Alex Zhang	Attendee
Mumbai	3/12/2024	National	Dr. Daniel	Attendee
Mumbai	3/12/2024	National	Dr. Daniel	Attendee
Mumbai	3/12/2024	National	Riya Singh	Presenter
Mumbai	3/12/2024	National	Riya Singh	Presenter
Tokyo	5/20/2024	International	Dr. Eva	Presenter
Tokyo	5/20/2024	International	Dr. Eva	Presenter
Tokyo	5/20/2024	International	Dr. Eva	Presenter
Tokyo	5/20/2024	International	Omar Khan	Presenter
Tokyo	5/20/2024	International	Omar Khan	Presenter
Tokyo	5/20/2024	International	Omar Khan	Presenter
Tokyo	5/20/2024	International	Dr. Alice	Panelist
Tokyo	5/20/2024	International	Dr. Alice	Panelist
Tokyo	5/20/2024	International	Dr. Alice	Panelist
Tokyo	5/20/2024	International	Dr. Eva	Presenter
Tokyo	5/20/2024	International	Dr. Eva	Presenter
Tokyo	5/20/2024	International	Dr. Eva	Presenter
Tokyo	5/20/2024	International	Omar Khan	Presenter
Tokyo	5/20/2024	International	Omar Khan	Presenter
Tokyo	5/20/2024	International	Omar Khan	Presenter
Tokyo	5/20/2024	International	Dr. Alice	Panelist
Tokyo	5/20/2024	International	Dr. Alice	Panelist
Tokyo	5/20/2024	International	Dr. Alice	Panelist
London	8/10/2024	International	Dr. Farah	Keynote Speaker
London	8/10/2024	International	Dr. Farah	Keynote Speaker
London	8/10/2024	International	Abdul Ahad	Presenter
London	8/10/2024	International	Abdul Ahad	Presenter
London	8/10/2024	International	Huzaifa Khan	Attendee
London	8/10/2024	International	Huzaifa Khan	Attendee
London	8/10/2024	International	Dr. Farah	Keynote Speaker
London	8/10/2024	International	Dr. Farah	Keynote Speaker
London	8/10/2024	International	Abdul Ahad	Presenter
London	8/10/2024	International	Abdul Ahad	Presenter
London	8/10/2024	International	Huzaifa Khan	Attendee
London	8/10/2024	International	Huzaifa Khan	Attendee

where Candidate key = [Project_Title, Faculty_Names, Student_Names, DOIs, Conference_Names]

2NF:

In the 1 normal form, many non-prime attributes depend on a part of the candidate key such as Project_objective, start_date and end_date depend on Project_Title and so on which is called partial dependency.

The table can be splitted to remove partial dependency.

Project

Project_ID	Project_Title	Project_Objective	Start_Date	End_Date	Funding_Amount	Funding_Body
1	AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024	50000	National Science Foundation
2	Quantum Computing Frameworks	Building scalable quantum algorithms	6/1/2021	12/31/2023	750000	Dept. of Science & Tech
3	Renewable Energy Microgrids	Efficient rural electrification using microgrids	1/1/2023	6/30/2025	600000	Green Energy Initiative
4	Human-Robot Interaction	Enhancing collaboration between humans and robots	9/15/2022	11/30/2024	400000	Tech Robotics Foundation
5	Language Tech for Education	Using NLP to support literacy in rural schools	5/1/2023	12/1/2025	350000	UNESCO

Primary Key = Project_ID

Faculty

Faculty_ID	Faculty_Name	Faculty_Email	Faculty_Department	Faculty_Research_Area
1	Dr.Ali	ali@uni.edu	CS	ML
2	Dr.Ahmed	ahmed@uni.edu	Biomedical	Medical Imaging
3	Dr. Claire	claire@uni.edu	Physics	Quantum Computing
4	Dr. Daniel	daniel@uni.edu	Electrical	Renewable Energy
5	Dr. Eva	eva@uni.edu	Robotics	HRI
6	Dr. Alice	alice@uni.edu	CS	Cognitive Systems
7	Dr. Farah	farah@uni.edu	Linguistics	NLP

Primary Key = Faculty_ID

Student

Student_ID	Student_Name	Student_Program	Student_Year
101	Hamza Malik	PhD	2
102	Asad Jamal	MSc	1
103	Alex Zhang	MSc	1
104	Riya Singh	PhD	3
105	Omar Khan	MSc	2
106	Abdul Ahad	PhD	2
107	Huzaifa Khan	MSc	1

Primary Key = Student_ID

Project_Faculty_Student

Project_ID	Faculty_ID	Student_ID	Student_Involvement_Hours_Per_Week
1	1	101	10
1	2	102	15
2	3	103	20
3	4	104	25
4	5	105	12
4	6	105	12
5	7	106	18
5	7	107	10

Composite Key = [Project_ID, Faculty_ID, Student_ID]. These three are Foreign keys as well.

Student_Supervisor

Student_ID	Student_Supervisor_ID
101	1
102	2
103	3
104	4
105	5
105	6
106	7
107	7

Candidate Key = [Student_ID, Student_Supervisor_ID]

Publication

DOIs	Publication_Titles	Publication_Years	Publication_types	Publishers
10.1000/xyz1	Deep learning for X-ray	2023	Conference Paper	Springer
10.1000/xyz2	ML in cardiology	2024	Book Chapter	IEEE
10.1000/xyz3	tum Entanglement in Logic	2023	Journal Article	Elsevier
10.1000/xyz4	Smart Microgrids for India	2024	Conference Paper	IEEE
10.1000/xyz5	Gesture Recognition in HRI	2024	Book Chapter	Springer
10.1000/xyz6	NLP for Primary Learning	2023	Journal Article	Wiley
10.1000/xyz7	Speech Tools for Kids	2024	Conference Paper	ACM

Primary Key = DOIs

Author

Author_ID	Authors_Names
101	Hamza
1	Dr. Ali
102	Asad
2	Dr. Ahmed
103	Alex Zhang
3	Dr. Claire
104	Riya Singh
4	Dr. Daniel
5	Dr. Eva
105	Omar Khan
6	Dr. Alice
106	Abdul Ahad
7	Dr.Farah
107	Huzaifa Khan

Primary Key = Author_ID

Publication_Author Details

DOI	Author_ID	Author_Order	Author_Type
10.1000/xyz1	101	1	Student
10.1000/xyz1	1	1	Faculty
10.1000/xyz2	102	1	Student
10.1000/xyz2	2	2	Faculty
10.1000/xyz3	103	1	Student
10.1000/xyz3	3	2	Faculty
10.1000/xyz4	104	1	Student
10.1000/xyz4	4	2	Faculty
10.1000/xyz5	5	1	Faculty
10.1000/xyz5	105	2	Student
10.1000/xyz5	6	3	Faculty
10.1000/xyz6	106	1	Student
10.1000/xyz6	7	2	Faculty
10.1000/xyz7	107	1	Student
10.1000/xyz7	7	2	Faculty

Candidate Key =[DOIs, Author_ID]

Conference

Conference_ID	Conference_Names	Conference_Locations	Conference_Dates	Conference_Types
1	Int.Conference on AI	Paris,France	6/15/2023	International
2	Nat. MedTech Meet	Islamabad, Pakistan	2/20/2024	National
3	Int. Quantum Symposium	Geneva	9/10/2023	International
4	Nat. Energy Innovation Summit	Mumbai	3/12/2024	National
5	Robotic Futures Conference	Tokyo	5/20/2024	International
6	EdTech Global Forum	London	8/10/2024	International
7	EdTech Global Forum	London	8/11/2024	International

continued

Conference_Status	DOIs
Active	10.1000/xyz1
Active	10.1000/xyz2
Active	10.1000/xyz3
Active	10.1000/xyz4
Active	10.1000/xyz5
Active	10.1000/xyz6
Active	10.1000/xyz7

Primary Key = Conference_ID, Foreign Key = DOIs

Conference_Participant Table

Conference_ID	Participants_Names	Participants_Roles
1	Dr.Ali	Keynote Speaker
1	Hamza	Presenter
1	Asad	Attendee
2	Dr.Ali	Keynote Speaker
2	Hamza	Presenter
2	Asad	Attendee
3	Dr. Claire	Presenter
3	Aex Zheng	Attendee
4	Dr. Daniel	Attendee
4	Riya Singh	Presenter
5	Dr. Eva	Presenter
5	Omar Khan	Presenter
5	Dr. Alice	Panelist
6	Dr. Farah	Keynote Speker
6	Abdul Ahad	Presenter
6	Huzaifa Khan	Attendee
7	Dr. Farah	Keynote Speker
7	Abdul Ahad	Presenter
7	Huzaifa Khan	Attendee

Composite Key = [Conference_ID, Participants_name]

3NF:

In the table PROJECT there is transitive dependency i.e. Project-> Funding body, Funding body -> Funding amount, which can be removed as,

Project

Project_ID	Project_Title	Project_Objective	Start_Date	End_Date	Funding_Body_ID
1	AI in Healthcare	Improving Diagnostic with AI	1/1/2020	9/12/2024	1
2	Quantum Computing Frameworks	ling scalable quantum algorit	6/1/2021	12/31/2023	2
3	Renewable Energy Microgrids	rural electrification using mi	1/1/2023	6/30/2025	3
4	Human-Robot Interaction	ollaboration between human	9/15/2022	11/30/2024	4
5	Language Tech for Education	LP to support literacy in rural	5/1/2023	12/1/2025	5

Primary Key = Project_ID, Foreign Key = Funding_Body_ID

Funding_Body

Funding_Body_ID	Funding_Body_Name	Funding_Amount
1	National Science Foundation	50000
2	Dept. of Science & Tech	750000
3	Green Energy Initiative	600000
4	Tech Robotics Foundation	400000
5	UNESCO	350000

Primary Key = Funding_Body_ID

Rest of the other tables are the same as in 2NF.

NO SQL PERSPECTIVE:

The Research & Conference Management System can also be effectively implemented using a NoSQL database, particularly in scenarios where data is semi-structured, grows rapidly, or requires high scalability and flexible schema design.
If the Research & Conference Management System is implemented using a NoSQL database like **MongoDB**, the design and storage of data would differ significantly from traditional to relational databases:

- **Flexible Data Structure**

  MongoDB allows storing documents without a fixed schema. This is ideal for academic records, which may vary widely across projects and researchers.
  Example:
  Some projects may have 2 faculty leads and 5 students, while others may have just 1 faculty and no students yet. A document structure in MongoDB easily accommodates both without altering the schema.'

- **Embedded Documents for Real-World Groupings**

  Instead of separate tables and foreign keys, related data can be embedded inside a single document.
  Example:
  A "Project" document could include an array of student contributors with fields like name, program, supervisor, and weekly involvement hours—all in one place.

- **Handling Authorship and Conference Roles**

  Academic systems require maintaining author order and participant roles for recognition.
  Example:
  A "Publication" document can store authors in an array like:
  "authors": [{"name": "Dr. Ali", "order": 1}, {"name": "Hamza Malik", "order": 2}]
  Similarly, a "Conference" document can list roles:
  "participants": [{"name": "Dr. Farah", "role": "Keynote Speaker"}]

- **Scalability and Performance**

  NoSQL supports horizontal scaling, making it suitable for large datasets like publication archives and multi-year research records.

  Example:
  If the university stores thousands of publications or tracks decades of project history, MongoDB scales easily without complex joins.

- **Reduced Complexity in Data Retrieval**

  Querying a document for all project-related information becomes simpler and faster.
  Example:
  Fetching a project with all its faculty, students, and publications requires just one query in MongoDB, instead of multiple joins in SQL.

CONCLUSION:

The proposed Research & Conference Management System is designed to streamline and centralize the management of research projects, academic publications, and conference participation within a university setting. By organizing data into a well-structured and normalized relational database, the system ensures data integrity, minimizes redundancy, and supports effective tracking of academic activities. It also establishes clear relationships between students, faculty, projects, publications, and conferences, enabling better academic visibility and recognition.

Furthermore, the design is adaptable for NoSQL implementation, offering flexibility for institutions that require scalability and performance optimization. Overall, the system lays a strong foundation for efficient academic data management and supports informed decision-making for research and academic growth.